

Assignment 6 (ELEC 341 L6_Stability)

Problem 1:

Tell how many roots of the following polynomial are in the right half-plane, in the left half-plane, and on the $j\omega$ -axis:

try -1

$$P(s) = s^5 + 3s^4 + 5s^3 + 4s^2 + s + 3$$

Use Routh-Hurwitz criterion.

s^5	1	5	1
s^4	3	4	3
s^3	$\frac{11}{3}$	0	0
s^2	4	3	0
s^1	$-\frac{11}{4}$	0	0
s^0	3	0	0

$$\frac{15-4}{3} = \frac{11}{3}$$

$$\frac{11 \cdot 3(4) - 3(0)}{11/3} = 4$$

$$\frac{11 \cdot 3(3) - 4(0)}{11/3} = 3$$

$$\frac{4(0) - \frac{11}{3}(3)}{4} = -\frac{11}{4}$$

$$\frac{-\frac{11}{4}(3) - (-\frac{11}{4})}{-\frac{11}{4}} = 3$$

$\downarrow$
 1
 3
 $\frac{11}{3}$
 4
 $-\frac{11}{4}$
 3

$\left. \begin{array}{l} +1 \text{ RHP} \\ +1 \text{ RHP} \end{array} \right\} \text{unstable.}$

2 roots on RHP.
 0 roots on $j\omega$ axis
 3 roots on LHP

No row of zeroes so
0 poles on $j\omega$ axis.